

Mecha Connect

Product Requirements Document (PRD)

Product Leadership Team

Version 1.1.0

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1 Document Metadata

1.1 Document Sign-offs

Role	Approver	Status	Date
Senior Product Manager	Lead PM	Approved	2026-07-24
Principal Engineer	Tech Lead	Approved	2026-07-24
Engineering Director	VP of Eng	Approved	2026-07-24
Senior UX Researcher	Design Lead	Approved	2026-07-24

1.2 Revision History

Version	Date	Author	Description
1.0.0	2026-07-20	Product Team	Initial Draft
1.1.0	2026-07-24	Review Panel	Added Epics, Dependency Map, Compliance, NFRs.

1.3 Glossary

TTR (Time-to-Resolution): Total time from user tapping "Help" to job completion.

Triage: Diagnosing the vehicle issue to determine the correct service type.

Provider/Mechanic: Independent contractor fulfilling the roadside assistance request.

Consumer: The stranded driver requesting help.

RAG: Retrieval-Augmented Generation AI architecture used to match consumer symptoms against verified repair manuals.

2 Executive Summary

2.1 Product Vision

Mecha Connect aims to be the most reliable, intelligent, and transparent roadside assistance platform globally. By combining Uber-style real-time dispatch infrastructure with an OpenAI-powered diagnostic engine, Mecha Connect transforms the historically opaque, high-anxiety experience of vehicle breakdowns into a seamless, deterministic, and safe digital interaction.

2.2 Problem Statement

- **For Customers:** Roadside assistance today is characterized by anxiety, unknown wait times, opaque pricing, and phone-based dispatch systems that lack context.
- **For Mechanics/Providers:** Independent mechanics struggle with customer acquisition, inefficient routing, delayed payments, and misdiagnosed call-outs.

2.3 Target Audience & User Personas

Persona 1: "Stranded Sarah" (Consumer)

28, non-technical. Needs fast response times, real-time tracking, upfront pricing, and AI assistance to diagnose "weird engine noises."

Persona 2: "Mobile Mechanic Mike" (Provider)

45, independent contractor. Needs accurate job descriptions, optimized routing, guaranteed digital payouts, and transparent platform fees.

3 Product Strategy & Objectives

3.1 North Star Metric

Time-to-Resolution (TTR): The total time elapsed from the user tapping "Help" to the mechanic completing the service. Target: < 45 minutes average.

3.2 Key Performance Indicators (KPIs)

KPI	Definition	Target (Q1)
Fulfillment Rate	% of requested jobs successfully completed	> 92%
AI Triage Accuracy	% of jobs where AI correctly predicts service	> 85%
Provider ETA Variance	Actual arrival time vs. System predicted ETA	± 5 minutes
App Crash Rate	Session crash rate (especially critical of-line)	< 0.1%
CAC (Customer)	Customer Acquisition Cost	< \$15

4 Product Scope (MoSCoW)

4.1 MVP (Must Have)

- User & Provider Onboarding (KYC/Stripe Identity).
- Profile & Vehicle Management (VIN lookup).
- One-Tap SOS Geolocation Request with Offline SMS fallback.
- AI Diagnostic Triage (Text/Audio input → Service Classification).
- Real-Time Algorithmic Matching & Dispatch.
- Live Map Tracking (Customer & Provider).
- Stripe Payment Integration (Pre-authorization & Payouts).
- Rating & Review System.

4.2 Fast Follows (Should Have)

In-app Voice/Text Chat, Provider In-App Wallet, Referral Program, Scheduled Maintenance Booking, Dynamic Surge Pricing.

4.3 Feature Dependency Map

Architectural Flow Sequence:

1. **User Auth & Profile** unlocks *Vehicle Management* and *Payment Setup*.
2. **Vehicle Management** feeds context into *AI Diagnostic Triage*.
3. **AI Triage** and **Payment Setup** feed into the *Dispatch Engine*.
4. **Dispatch Engine** triggers *Provider Matching*.
5. **Provider Matching** enables *Live Map Tracking*.
6. **Job Completion** triggers *Payment Capture*, *Provider Wallet Payouts*, and *Reviews*.

5 User Journeys

5.1 Core Request Flow (Customer to Provider)

1. **Customer** taps "I Need Help" in Mobile App.
2. **App** prompts for issue description (Voice/Text).
3. **Customer** inputs: "My car won't start, clicking noise."
4. **App** sends symptom text + Vehicle metadata to **AI Triage**.
5. **AI** classifies as "Dead Battery", recommends "Jump Start".
6. **App** confirms Jump Start and shows Upfront Price (\$65).
7. **Customer** authorizes Stripe Pre-Auth.
8. **Dispatch Engine** broadcasts request to available mechanics (10km radius).
9. **Provider** accepts Job.
10. **Dispatch Engine** confirms match, sends ETA and Live Map to Customer.
11. **Provider** arrives, completes job, and marks as complete.
12. **System** finalizes charge, routes payout, and prompts for reviews.

6 Core Features & Detailed Requirements

6.1 Epic 1: Intelligent SOS & Triage

Feature 1.1: AI Diagnostic Chatbot

Translate user anxiety/symptoms into actionable service enums.

- **Requirements:** Accept text/speech. Map LLM output to standard enum. Pass vehicle metadata as strict context.
- **Business Rules:** AI must NEVER guarantee a price. Trigger 911 overlay if words like "accident" or "fire" are detected.
- **Acceptance Criteria:** Given user inputs "hissing from tire", AI selects "Flat Tire Repair".

Feature 1.2: Geolocation & Offline Fallback

Capture exact coordinates in poor network conditions. If offline, generate compressed SMS payload to a Twilio fallback number.

6.2 Epic 2: Real-Time Dispatch & Matching

Feature 2.1: Algorithmic Routing

Broadcast to nearest 3 providers. Weighting: Distance (60%), Provider Rating (30%), Completion Rate (10%). Expand radius by 5km if no acceptance in 45 seconds.

6.3 Epic 3: Payments & Platform Economics

Feature 3.1: Upfront Pricing & Pre-Authorization

- **Rules:** Base Rate + (Per Km * Distance) + Surge. Platform retains 20%, Provider receives 80%. MUST pre-auth 110% of estimated price.
- **Cancellation:** < 5 mins = Free. > 5 mins = \$25 fee routed to provider.

6.4 Epic 4, 5, 6 & 7 Summaries

- **Epic 4 (Vehicles):** NHTSA VIN API integration to auto-populate vehicle specs (critical for EV dispatch).
- **Epic 5 (Mechanic Portal):** Stripe Connect integration for 2-day rolling payouts or Instant Payouts (1.5% fee).
- **Epic 6 (Trust):** Mandatory dual-blind 1-5 star ratings before accepting next job.
- **Epic 7 (Admin):** "God-Mode" dashboard for internal Ops to force-assign mechanics or force-refund payments via Stripe.

7 Analytics & Telemetry

Core Mixpanel/Amplitude funnel events:

- app_opened, sos_initiated
- ai_triage_completed (Props: predicted_service, confidence_score)
- payment_authorized, job_dispatched, job_accepted
- job_completed (Props: final_price, rating)
- AI Telemetry: ai_fallback_triggered, user_corrected_ai

8 Non-Functional Requirements (NFRs)

8.1 Performance & Scalability

- **API Latency:** P95 latency for critical paths must be < 500ms.
- **Concurrency:** Support 10,000 concurrent active jobs regionally.
- **Cold Start:** App loads in < 2.0 seconds on mid-tier Androids.

8.2 Security, Compliance & Accessibility

- **PCI-DSS:** SAQ-A scope via Stripe Elements (no raw card data touches our servers).
- **GDPR/CCPA:** Right to be Forgotten self-serve toggle.
- **Accessibility:** WCAG 2.1 AA Compliance (Screen-reader support critical for visually impaired users in emergencies).
- **Disaster Recovery:** RPO < 5 minutes, RTO < 1 hour.

9 Launch Strategy & Go-To-Market

- **Phase 1 (Alpha):** Internal dogfooding and simulated breakdowns.
- **Phase 2 (Closed Beta):** Single geography (e.g., Austin, TX). Cap at 50 vetted providers.
- **Success Criteria for V1:** Load test sign-off (5,000 concurrent), Zero P0/P1 security vulnerabilities, 100 verified active providers in launch city.